

Adoption of AI Systems for Circular Construction Procurement

A Comparative Analysis of the UK and Türkiye

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Integrating circular procurement (CP) practices in the construction industry is vital to advancing sustainability and minimising the environmental impact of construction and demolition waste (CDW). Despite the substantial benefits, the implementation of circularity in procurement processes remains limited, particularly in terms of adopting artificial intelligence (AI) in the construction sector. The primary aim of this study is to explore the role of AI in supporting the implementation of circular procurement in the construction sector, with a specific focus on applications in the United Kingdom (UK) and Türkiye. The key objectives are to identify how AI technologies are currently applied to enhance circular procurement practices and to evaluate the opportunities and constraints associated with AI adoption by examining existing policy frameworks, data infrastructures, and technical capacities. Through a systematic literature review, the study identifies six key themes: circular economy policy and strategy, circular material design, circular construction, supply chain management, CDW management, and resource recovery. Based on the established themes, a comparative analysis is conducted between the UK and Türkiye, examining their positions regarding AI integration, highlighting overlapping patterns and unique challenges. AI presents considerable potential for advancing circular procurement, particularly through the development of data-driven decision-making, effective management protocols, resource optimisation, and improved supply chain management. The challenges encompass a lack of data for CDW monitoring, insufficient legislative frameworks, high upfront costs, a shortage of technical expertise, and ethical concerns about accountability. The findings emphasise the need for AI platforms that facilitate the adoption of circular construction practices, supporting the development of dynamic, responsive, and smart procurement strategies.

Keywords: artificial intelligence, circular economy, circular procurement, AI-driven construction procurement, construction and demolition waste, resource optimisation

INTRODUCTION

Artificial intelligence, along with its various domains, such as robotics, machine learning, optimisation, knowledge-based systems, and extended reality, has driven significant advancements and transformative changes in various fields of the construction industry (Balogun *et al.*, 2024). One critical area where AI can play a pivotal role is in enabling Circular Economy (CE) principles in construction. Circular economy envisions a regenerative system that optimises the use of materials and retains their value throughout their lifecycle (Bertino *et al.*, 2021). CE is necessary for enabling resource efficiency, energy conservation, waste minimisation, and sustainability within the construction industry (Sajid, Aftab and Ullah, 2024).

Several principles strongly characterise the CE at the building level and are classifiable into three main groups: (1) design process aimed at adaptability and reversibility, (2) resource and waste management aimed at reuse and recycling, and, (3) business models aimed at extending the life and value of products (Giorgi, Lavagna and Campioli, 2019). Among these classifications, prioritising waste prevention as a first goal for the efficient and effective use of resources is essential. This can be accomplished by extending the building's lifespan, enhancing its durability, and prioritising maintenance, repair, and reuse.

Circular Procurement (CP) management plays a significant role in advancing CE objectives (Qazi and Appolloni, 2022). CP is defined as the process by which organisations purchase works, goods, or services that aim to contribute to closed energy and material loops within supply chains while minimising and avoiding negative environmental impacts and waste creation across their lifecycle. Instead of following the traditional models rooted in the linear *take-make-waste* paradigm, CP focuses on extending the life cycle of products, encouraging recycling, reducing environmental impact, and enabling energy conservation by

allowing material reuse at the end of the lifecycle (Ruiz, Ramon and Domingo, 2020).

The principles of CE have gained traction in recent years as a means to address the construction industry's significant environmental impact. However, critical studies documenting the barriers to adopting CP within the construction industry remain scarce. Besides, the construction industry has been relatively slow to adopt digitalisation, with a discernible gap in integrating AI across the various phases of the construction project lifecycle (Abioye *et al.*, 2021). The practical implementation of CE concepts within the construction sector has limited the potential for leveraging AI and other digital technologies to enable more comprehensive circular business models and material flows (Adams, 2017). This paper presents a comprehensive review of the challenges and opportunities of adopting AI systems to promote circularity in construction procurement, with a specific focus on its application in the United Kingdom (UK) and Türkiye. The following section includes background research in AI and circular construction systems.

RESEARCH BACKGROUND

The fundamental benefit of AI is creating a platform for decision support through valuable insights and results collected from large and complex data sets and compressed into a manageable form. This creates an opportunity to enhance the solutions to prevailing business problems for industrial manufacturers and their customers (Sjödin, Parida and Kohtamäki, 2023). These solutions develop new classes of circular business models that leverage the unique AI capacities of augmentation (e.g., optimisation solutions) and automation (e.g., autonomous solutions) (Raisch and Krakowski, 2021).

Utilising AI to support circular construction services includes optimising efficiency, reducing waste, and improving sustainability while enabling smart decision-making and predictive

maintenance (Kristoffersen *et al.*, 2020). The literature on efficiency optimisation focuses on enhancing operational processes and resource allocation. (Chen *et al.*, 2023; Song *et al.*, 2024). On the other hand, construction waste reduction emphasises the intelligent management of Construction and Demolition Waste (CDW). Several studies have applied optimisation models to estimate the recoverable values of construction materials, incorporating cost, duration, and environmental impact (Mollaei *et al.*, 2023). Complementary frameworks based on Building Information Modelling (BIM) facilitate the specification of recyclable or reusable materials, and machine learning applications have enhanced waste management strategies (Balogun *et al.*, 2024). Additionally, studies have highlighted the role of AI in reducing carbon emissions, supporting net-zero targets, and facilitating lifecycle assessments (Adewale *et al.*, 2024; Li *et al.*, 2024).

Most research on AI in circular construction focuses on enhancing design and construction practices, optimising resource utilisation, and promoting sustainability, whereas research on construction procurement operations remains limited. The adoption of CP effectively integrates procurement practices with the principles of circular objectives. The CP process emphasises evaluating the design, production, sale, reuse, and recycling of products to identify ways to extract and maximise their value both during use and at the end of their life. CP practices are crucial for embedding circularity throughout the entire construction value chain, including conventional methods, resistance to change, and inadequate integration of digital tools. The next section outlines the methodology for structuring themes in the current state of AI applications in CP, as well as future research directions.

METHODOLOGY

A Systematic Literature Review (SLR) was conducted to select relevant research associated

with the adoption of AI in CP within the construction industry using the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol (Page *et al.*, 2021). To organise the selected studies, a semi-automated classification method was employed based on density maps and term proximity derived from VOSviewer analysis. This analysis facilitated the identification of AI themes used in CP and the clustering of selected research according to these AI themes. Furthermore, a comparative analysis was carried out on the research data from the UK and Türkiye to explore how AI is integrated into construction procurement circularity.

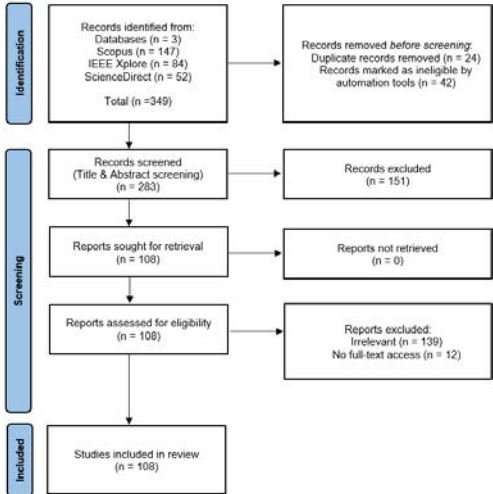
Systematic review of AI studies in CP

The SLR process involves evaluating keyword-based articles retrieved from Scopus, IEEE Xplore, and ScienceDirect databases. The eligibility criteria require articles that incorporate the specified keywords within their titles, abstracts, or keyword sections. In the identification stage, a systematic search was conducted using the following Boolean query:

("Circular Procurement" OR "Sustainable Procurement" OR "Circular Economy" OR "Circular Construction" OR "Circularity") AND ("Construction" OR "Built Environment" OR "Building" OR "Infrastructure") AND ("Artificial Intelligence" OR "AI" OR "Machine Learning" OR "Deep Learning" OR "Data-driven" OR "Automation")

In thematic focus, the search was refined by eliminating subject areas unrelated to construction, such as energy, social sciences, business, economics, and medicine. The initial search retrieved a total of 349 records, including all document types, and was reduced to 283 records by limiting the study to original articles, review papers, and conference papers (Scopus: 147, IEEE Xplore: 84, ScienceDirect: 52). These records were analysed for duplicates, as studies appeared in multiple databases. After removing 42 duplicates, the remaining 259 records underwent title and abstract screening to exclude

articles that were clearly outside the scope of this study. A final dataset of 108 documents was retrieved for full-text review thematic analysis. The PRISMA flow diagram is presented in Figure 1.



Thematic mapping of AI studies in CP

The AI theme clustering analysis of the selected articles was conducted using a semi-automated classification based on the term frequency and link strength analysis. VOSviewer was used to analyse and visualise clusters of keyword occurrence in the dataset to understand density maps and term proximity (Figure 2).

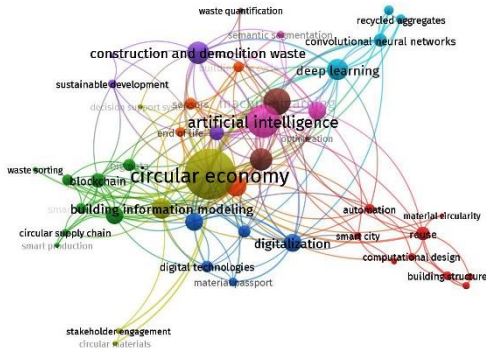


Figure 1
PRISMA Flow
Diagram

Following the identification of keyword clusters, a rule-based Python script was developed to classify the retrieved articles into thematic groups. In the clustering phase, metadata from the SLR was utilised, with a focus on titles and abstracts, to categorise theme-specific dictionaries. Using frequency scores for each research record, the six distinct themes were identified as: (1) AI for CE policy and strategy; (2) AI for circular materials and product design; (3) AI in circular construction and buildings; (4) AI in circular supply chains and manufacturing; (5) AI in CDW management; (6) AI in waste management and resource recovery. The distribution of six themes across 108 research records is illustrated in Figure 3.

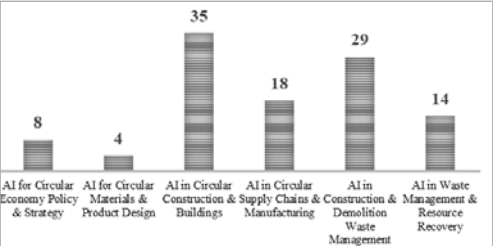


Figure 3
Distribution of
studies across AI
themes

Each theme was operationalised through a dictionary of relevant keywords representing key concepts (Table 1). The first theme encompasses studies that provide an overview of the strategy, adoption, and policy implications of AI in developing decision-making or assessing the impact of CE initiatives. This includes data-driven policy analyses such as the “zero waste city” policy (Bi *et al.*, 2024), lifecycle-oriented AI frameworks (Yannou *et al.*, 2024), diffusion models that predict the adoption of CE (Oluleye, Chan and Saka, 2024), and the use of AI for sustainability and CE (Salimi and Taherkhani, 2024).

The circular materials and product design theme includes the use of AI to optimise material properties and construction products. Studies in this category explore AI-driven models that enhance the performance of materials through machine learning optimisation (Kazemi and

Figure 2
Network
visualisation of
keyword co-
occurrence

Mirjalili, 2024; Ranjith *et al.*, 2024), deep learning to classify mixed-material inputs (Chen and Yu, 2023), and the use of AI in assessing reclaimed materials (Pasca *et al.*, 2025).

Table 1
Theme-specific
keywords

AI Theme	Theme-specific keywords
CE policy & strategy	"policy", "adoption", "zero waste", "diffusion", "literature review", "decision making", "data-driven policy", "regulation", "stakeholder"
Circular material & product design	"recycled concrete", "material design", "glass", "timber", "mixtures", "mechanical properties", "industrial by-products", "product design"
Circular construction & buildings	"building information modelling", "BIM", "digital twin", "design", "circular building", "reuse potential", "component prediction", "maintenance optimization", "circular construction"
Circular supply chain & manufacturing	"supply chain", "manufacturing", "blockchain", "IoT", "digital product passport", "traceability", "industry 4.0", "production planning", "forecasting"
CDW management	"demolition", "CDW", "construction waste", "waste classification", "deep learning", "waste prediction", "waste logistics", "material recovery facility", "sorting"
Waste management & resource recovery	"municipal waste", "recycling", "robotics", "CNN", "material recovery", "sorting", "resource recovery", "incineration", "wastewater"

The circular construction studies address how AI supports circularity in design, reuse prediction, and operational optimisation. Key studies highlight optimisation of deconstruction protocols (Kunic *et al.*, 2024), deep learning for component identification (Lux *et al.*, 2023) and the role of AI in adaptable building systems (Williams *et al.*, 2023). Circular supply chains and manufacturing studies examine the application of AI in enhancing supply chain visibility, traceability, and circular logistics. The literature discusses blockchain-based traceability tools and product passports (Elghaish *et al.*, 2023), predictive logistics (Danish and Senjyu, 2023), reverse logistics models supported by machine learning (Farshadfar *et al.*, 2025), and industry 4.0 and digital manufacturing systems for reuse design (Akbari, 2023).

CDW management research focuses on AI-powered solutions for minimising CDW through sorting, classification, and forecasting. Computer vision and deep learning methods are used for waste recognition and robotic separation (Lin *et al.*, 2022; Dodamegama *et al.*, 2024). Additionally, remote sensing and 3D scanning are being used for material identification (Kronenwett *et al.*, 2024), demolition waste forecasting (Akanbi

et al., 2020), and hazardous materials detection during waste audits (Wu *et al.*, 2022). The last theme encompasses broader applications of AI in municipal and industrial waste systems. The examples include robots using computer vision to facilitate sorting at high-speed municipal recycling centres (Cosoli *et al.*, 2024), the municipal-level integration of AI and IoT (Lakhout, 2025), smart landfill management (Solainayagi, 2024), and wastewater treatment optimisation (Singh, Chakraborty, and Sehgal, 2023).

COMPARATIVE ANALYSIS OF THE UK AND TÜRKİYE FOR AI INTEGRATION IN CIRCULAR PROCUREMENT

This section provides a comparative analysis of how the UK and Türkiye have integrated AI into circular construction procurement, with a focus on thematic applications and real-world feasibility. Building upon six AI application clusters, operational-scale deployments, innovation capacities, and socio-technical readiness of both countries were assessed.

Opportunities and emerging research trends in AI-enabled CP

The integration of AI into CE reflects diverse policy and implementation ecosystems. In the UK, the CE policy is supported by governance frameworks, such as the Ethics Guidelines for Trustworthy AI, AI adoption for sustainability, and the AI White Paper, which promote AI-driven decision-making in construction. Additionally, initiatives such as the Resource and Waste Strategy and the Centre for Digital Built Britain aim to develop waste management policy and enhance construction procurement. The circular construction approach has been enhanced through pilot projects, including the use of AI in BIM, digital twins for lifecycle optimisation and material reuse, generative design, and BIM-integrated material passports. Manufacturing integration is supported by industry-government

platforms such as UKRI and YardLink that enhance supply chain visibility and material sourcing, streamlining procurement and connecting contractors with suppliers. In CDW management, AI-robotic systems, such as ZenRobotics and Recycleye, are utilised for automated sorting and pre-demolition audits, leveraging advanced image recognition. Along with AI-enabled material recovery tools, AI in waste management is recognised for transforming operational efficiency in waste recycling.

In Türkiye, the National AI Strategy identifies CE in the built environment as a priority, and the DEEP project demonstrates increased institutional involvement in CE. For circular material design, early studies have been conducted on the use of recycled materials and design for disassembly practices (Esin and Cosgun, 2007). A notable example includes the Ekolojim prototype developed by Atatürk University for on-site waste separation. Additionally, AI-assisted sorting device prototypes and smart city projects show the increasing involvement of Türkiye in CE. Edge-AI drone-based surveillance system prototypes developed with TUBITAK support provide real-time monitoring capabilities on construction sites. Regarding the circular supply chain, a Turkish small and medium-sized enterprise (SME) has launched an AI-powered textile waste platform that offers digital product passports toward circular manufacturing and traceability. A Türkiye-based digital platform example is the inSuppliers, which connects construction stakeholders and promotes circular procurement by integrating sustainability filters and verified supplier data into early-stage decision-making. Initiatives supported by the Zero Waste project, including end-to-end waste management solutions and reverse vending machines, present opportunities for AI-based waste management (Telit, 2024). Besides, the regulatory shift aligned with licensed recyclers and registered collectors aims to modernise waste management and improve traceability and recycling rates.

Challenges and research gaps in AI-enabled CP

In both countries, AI implementation for CP is constrained by infrastructural and socio-technical factors. In the UK, despite strong policy support, the primary challenge is striking a balance between innovation and accountability in AI systems. Unregulated AI could give unfair advantages or pose safety risks, which require policies to maintain trust (Abioye *et al.*, 2021). A 2025 UK parliamentary committee report revealed that over 60% of government bodies cited poor data quality, and 70% reported AI talent shortages as key barriers to AI adoption in the public sector. Additionally, many SMEs face significant challenges in covering the high upfront costs of implementing AI-driven technologies in construction and waste management (Fang *et al.*, 2023). Therefore, concerns about safety, accountability, and trust continue to shape the regulatory debate. Additionally, technical integration issues, such as a lack of interoperability between AI platforms and fragmented data systems, limit cross-sector AI adoption, particularly in supply chains (Kristoffersen *et al.*, 2020).

Turkey currently lacks a dedicated legislative framework to regulate AI, with a proposal for AI legislation presented in June 2024. Unlike the EU AI Act, the proposed Turkish AI law proposal lacks classifications, only indicates special measures for high-risk systems, and does not have clear definitions for high-risk AI systems (Doğan Kösterit, 2025). Besides, the regulatory enforcement in Türkiye's construction waste strategy is inconsistent, and the reporting of CDW data and accountability are unclear (Gözübenli, 2025). There is also a lack of localised datasets for AI training, limiting the performance of imported AI tools that are not adapted to Türkiye's material profiles and construction standards (Salihoglu *et al.*, 2021). The lack of structured and local datasets, particularly for a significant number of recyclable materials, presents a role for informal

waste pickers, whose contributions are statistically underestimated. This requires replicable methodologies for enumerating informal waste actors and emphasising the need for community-based participatory approaches to ensure inclusive data integration and visibility (Cook *et al*, 2024).

CONCLUSION

This study examines how AI can facilitate circular practices in construction procurement, with a comparative focus on the UK and Türkiye. The systematic review categorised AI applications into six thematic areas, providing a foundation for a comparative analysis of policy readiness, technological adoption, institutional capacity, and practical applications. This analysis highlights trends and challenges in both countries for advancing AI-enabled circularity in the construction sector. The findings emphasise the need for designed AI platforms that foster the adoption of CP practices to support collaborative design and the development of smart procurement systems with real-time data integration across stakeholders. Besides, it is revealed that while the UK exhibits higher institutional readiness, supported by clear policy frameworks and active pilot programs, practical implementation remains fragmented. Conversely, Türkiye demonstrates innovation at the prototype level, particularly through academic and SME initiatives; however, it faces barriers due to a weak regulatory infrastructure and limited access to data frameworks that account for the contributions of the informal sector. Leveraging participatory methods can help inform policy and support the design of AI-compatible circular systems. Overall, in both contexts, the socio-technical dimensions of AI adoption, including labour transitions, data sovereignty, and institutional trust, require further investigation.

This study contributes to the field by proposing a comparative approach that connects policy-level discourse with operational feasibility

through case studies and pilot initiatives. However, the review is limited to published documents in English and may not reflect undocumented or non-digitised initiatives. Furthermore, the analysis does not provide a quantitative assessment of the effectiveness or maturity of the discussed AI applications. The findings can inform policymakers seeking to enhance AI readiness in the construction sector, particularly in emerging economies. Industry stakeholders can also use the mapped opportunities and barriers to develop targeted procurement tools, training programmes, and open-data strategies. Lastly, researchers can extend the framework to other countries or sectors undergoing circular transitions.

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